

U.S.N.

B.M.S. College of Engineering, Bengaluru-560019

Autonomous Institute Affiliated to VTU

June 2025 Semester End Main Examinations**Programme: B.E.****Semester: VI****Branch: CS(DS)/AIDS****Duration: 3 hrs.****Course Code: 23DS6PCNGD****Max Marks: 100****Course: Next Gen Databases**

Instructions: 1. Answer any FIVE full questions, choosing one full question from each unit.
2. Missing data, if any, may be suitably assumed.

Important Note: Completing your answers, compulsorily draw diagonal cross lines on the remaining blank pages. Revealing of identification, appeal to evaluator will be treated as malpractice.			UNIT - I	CO	PO	Marks
	1	a)	Company ABC is planning to build a cost-effective and scalable data analytics platform to handle rapidly growing volumes of semi-structured data. Apply your understanding of Hadoop's architecture and core features to explain how Hadoop can effectively address these requirements.	CO1	PO1	08
		b)	You are developing a large-scale application that needs to access and analyze vast amounts of user behavior data. Using a diagram, illustrate how the different components of the Google software stack can be integrated to support this process. Clearly show the interaction between your application and each layer of the stack.	CO1	PO1	08
		c)	Describe the second database revolution with relevant examples.	CO1	PO1	04
			OR			
	2	a)	A user submits an HQL query for processing. Using an architectural diagram, illustrate how Hive interacts with other components to execute the query. Clearly depict the flow of control and data among the components involved.	CO1	PO1	08
		b)	Analyze how the YARN architecture in Hadoop 2.0 facilitates efficient resource allocation and workload isolation, and how it can be utilized to manage and scale multiple concurrent machine learning workloads in a distributed computing environment.	CO1	PO1	08
		c)	Discuss the major drawbacks of using sharding as a strategy to scale relational databases for large-scale web applications	CO1	PO1	04
			UNIT - II			
	3	a)	Create a database for a library management system using PostgreSQL with the given tables. Authors (author_id, name, birth_year, county) Books (book_id, title, author_id, category, published_year, copies_available) Members (member_id, name, email, membership_date) Borrowings (borrow_id, book_id, member_id, borrowed_date, return_date)	CO3	PO3	10

		Write PostgreSQL to perform the following operations on the above table. 1. Create the above tables with the primary and foreign key constraints. 2. Find the author with the least book in the library. 3. Find the members who never borrowed a book. 4. Find the total number of authors who have at least one book borrowed in 2023. 5. Create a view that shows all books along with the author's name.																												
	b)	Assume you are developing an inventory management system for a warehouse using PostgreSQL. Each time a new item is added to the products table, you want to automatically record this activity in a product_audit table for inventory tracking. Design the necessary tables and implement a PostgreSQL trigger that captures: the product ID, the operation type, and the timestamp of the action. Also discuss the advantage of using triggers and explain how triggers help for the above scenario.	CO3	PO3	10																									
		OR																												
4	a)	Write the PostgreSQL query to create the given two tables and demonstrate how to perform the following Inner Join, Left Join, Right Join and Full Outer Join. Provide the PostgreSQL syntax for each join and explain the difference in the results returned by each type. <table><tr><th colspan="2">Student Table</th><th colspan="3">Course Table</th></tr><tr><th>S_Id</th><th>Name</th><th>C_ID</th><th>Course_Name</th><th>S_Id</th></tr><tr><td>1</td><td>Alice</td><td>101</td><td>Math</td><td>1</td></tr><tr><td>2</td><td>Bob</td><td>102</td><td>Science</td><td>2</td></tr><tr><td>3</td><td>Charlie</td><td>103</td><td>History</td><td>4</td></tr></table>	Student Table		Course Table			S_Id	Name	C_ID	Course_Name	S_Id	1	Alice	101	Math	1	2	Bob	102	Science	2	3	Charlie	103	History	4	CO3	PO3	10
Student Table		Course Table																												
S_Id	Name	C_ID	Course_Name	S_Id																										
1	Alice	101	Math	1																										
2	Bob	102	Science	2																										
3	Charlie	103	History	4																										
	b)	Write a PostgreSQL full-text search query that finds titles containing both “wild” and “call” (or any stemmed versions) and explain the execution and output of the query for the given table with attributes Id and Title. <table><tr><th>Id</th><th>Title</th></tr><tr><td>1</td><td>The call of the wild</td></tr><tr><td>2</td><td>Wild Adventures in the jungle</td></tr><tr><td>3</td><td>Calling all wild creatures</td></tr><tr><td>4</td><td>The wilderness and the call</td></tr></table>	Id	Title	1	The call of the wild	2	Wild Adventures in the jungle	3	Calling all wild creatures	4	The wilderness and the call	CO3	PO3	06															
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1	The call of the wild																													
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	c)	Differentiate between group by and window function in PostgreSQL.	CO2	PO2	04																									
		UNIT - III																												
5	a)	The book details are stored in a CouchDB database. Each document has the following structure: { "_id": "b1", "type": "book",	CO3	PO3	10																									

		<pre>"title": "Clean Code", "author": "Robert C. Martin", "genre": "Software Engineering", "year": 2008, "price": 35.0 }</pre> i) Write a view that emits book titles by genre. ii) Write a view to display all the details of books published after the year 2000. iii) Create a map-reduce view to calculate the total price of all books. iv) Create a map-reduce view to calculate the total price of books per genre. v) Create a view that retrieves all books whose price falls within a specific range.			
	b)	Imagine you are building an e-commerce platform. How could a document-oriented CouchDB database be used to store and manage product information efficiently? Provide an example schema for storing product details, including variants (like size and color), and explain the advantages of using a document model in this case.	CO2	PO2	10
		OR			
6	a)	Design a Library Management System using CouchDB, a NoSQL document-oriented database. Use appropriate curl commands to perform the following operations: i. Create Library databases for books and members in CouchDB. ii. Insert two new books and two new members into their respective databases. iii. Retrieve all books authored by "J.Row". iv. Update the title of a specific book using its document ID. v. Delete a member record from the database.	CO3	PO3	10
	b)	Discuss the Strength and Weakness of CouchDB database with examples.	CO1	PO1	10
		UNIT - IV			
7	a)	Using Gremlin, perform the following operations and display the output of each query: i) Initialize the Graph ii) Add two person vertices—"Alice" (age 30) and "Bob" (age 35)—and one software vertex—"Gremlin" iii) Connect each person to the "Gremlin" software with an edge labeled uses iv) Write a traversal that returns every vertex's ID, label, and properties. v) Find the names of all persons who uses the software named "Gremlin".	CO3	PO3	10
	b)	Demonstrate with examples how the following properties are achieved in Neo4J database : i) ACID Property ii) High Availability iii) Backup	CO1	PO1	10

			OR			
	8	a)	Create a course enrollment database at a university. Create two node types: Student (studentID, name), Course (courseID, title) and one relationship type ENROLLED_IN (from Student → Course). Write Cypher query to perform the following operations: i) Create the three Nodes each and relationship. ii) Display All Students. iii) List Courses for “Jane”. iv) Order All Courses by Title (Ascending). v) Find Students Enrolled in BOTH “History 201” and “Computer Science 101”	CO3	PO3	10
		b)	Design a Neo4j cluster using three database instances (one master and two slave) with separate cluster coordinator nodes.	CO1	PO1	06
		c)	Discuss the weakness of Neo4J database.	CO1	PO1	04
			UNIT - V			
	9	a)	You are building a Redis-based content aggregator that classifies websites into various categories. You have three sets of URLs under the following keys: news, tech and education Using Redis Set commands , perform the following operations and display the output. i) Add the following URLs to the respective categories using the appropriate command: news: cnn.com, nytimes.com tech: nytimes.com, github.com, apple.com education: udemy.com, github.com ii) Find the websites that are common to both tech and education. iii) Identify which news websites are not part of the tech category. iv) Combine all websites from news, tech, and education into a new set called all_sites. v) Remove nytimes.com from the tech set (it was miscategorized).	CO3	PO3	10
		b)	A social media platform uses Redis to cache user profiles and timelines. Illustrate with a diagram how Redis architecture ensures fast access, crash recovery, and load distribution in such a system explaining each component in detail.	CO2	PO2	10
			OR			
	10	a)	Design an online e-commerce platform using a polyglot persistence strategy. For each major feature, specify which database technology you would use and justify your choice.	CO2	PO2	08
		b)	Differentiate between the Blocking Queue and Publish-Subscribe messaging patterns in Redis.	CO1	PO1	08
		c)	Using the Redis Hash commands, perform the following operations on the hash at key user:alice. i) Store Alice’s full name, email and initial login count (0) in one command. ii) Retrieve only the email field from Alice’s profile. iii) Increment Alice's login_count by 1. iv) Retrieve All Fields in Alice’s Profile	CO2	PO2	04
